



Certificate

We hereby confirm that

Nithin Balasubramanian

has completed the intensive program

Data Science & AI

with 960 hours of programming practice (corresponds to 720 lessons) successfully.
The course has been delivered remotely.

Participation from 27.04.2026 until 26.08.2026

Franz Wilhelmy
Program Manager

Petra Kischkewitz
Managing Director

Tolga Sen
Lead Coach

Curriculum Data Science, Machine Learning & AI

In 960 lessons the students have learned, discussed and practiced the following content in several tasks and projects:

Programming & Databases

Python
○ Pandas
○ NumPy
○ Scikit-Learn
○ Statsmodels
○ TensorFlow / Keras

Visualization
○ Matplotlib
○ Seaborn
○ Plotly

Unix / Command Line

Git / GitHub

Basic SQL

Exploratory Data Analysis (EDA) & Statistics

Data Cleaning & Preparation

Data Analysis
○ Distributional Analysis
○ General Statistics

Data Visualization
○ Distribution Plots
○ Relationship Plots
○ Geo Visualizations

Feature Engineering

Machine Learning Algorithms

Supervised Learning
○ Linear Regression
○ Logistic Regression
○ Decision Tree
○ KNN
○ Ensemble Methods

Unsupervised Learning
○ K-Means
○ Clustering
○ Dimensionality Reduction
○ Recommender Systems

Model Tuning & Optimization

Bias-Variance Tradeoff
Regularization
Cross-Validation
Hyperparameter Tuning
Gradient Descent
Cost Functions

Evaluation & Performance Metrics

Regression & Classification Metrics
Confusion Matrix
Error Analysis

Deep Learning

Artificial Neural Networks
Convolutional Neural Networks
Pre-trained Networks & Transfer Learning

Natural Language Processing

LLM
Prompt Engineering
RAG
AI Agents

Time Series Analysis

APIs & Web Scraping

Streamlit

Portfolio Projects

Project 1: Exploratory Data Analysis
Project 2: Predictive Modeling
Capstone Project (4 weeks)

Collaborative Working & Social Learning

Pair Coding
Agile Workflow
Self-organization Skills
Group Work, Individual Exercises
Communication with Stakeholders &
Business Presentations
Career Coaching & Mentoring